



DYNAMIC MODELLING AND SIMULATION OF A CONVEYER BELT UNDER VARIABLE LOAD CONDITIONS

Team: [Sushant Shetty , Ishaan Sheth]

Course: Dynamic System Modeling (702MH0C023)

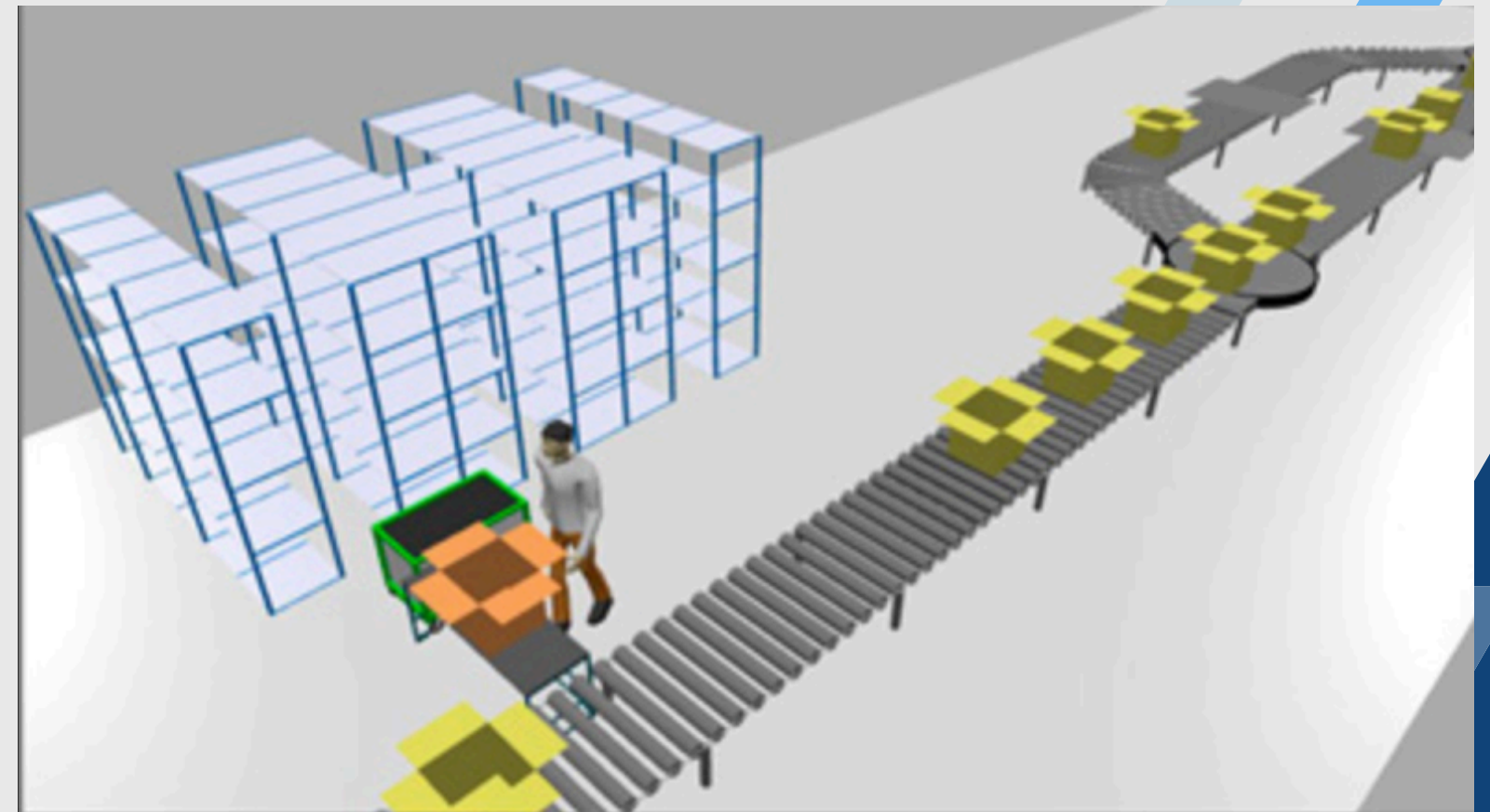
B.Tech Mechatronics, Sem IV | NMIMS MPSTME Mumbai

Category: [A/B/C/D/E] | Complexity: [Medium/High/Very High]

COs Exercised: [CO1, CO2, CO3, CO4, CO5] | Date: [DD/MM/YYYY]

SYSTEM INTRODUCTION & MOTIVATION

- What is the system? A motorized conveyor belt transporting a block load governed by Coulomb friction — belt moves at constant 1.5 m/s, block placed on it from rest
- Why does it matter? Variable loads cause speed mismatches, friction-driven acceleration, and positional drift — critical in manufacturing, mining, and baggage handling systems
- PBL Problem Statement: How does a block's velocity and position respond when placed on a moving conveyor belt with dry Coulomb friction under variable load conditions?



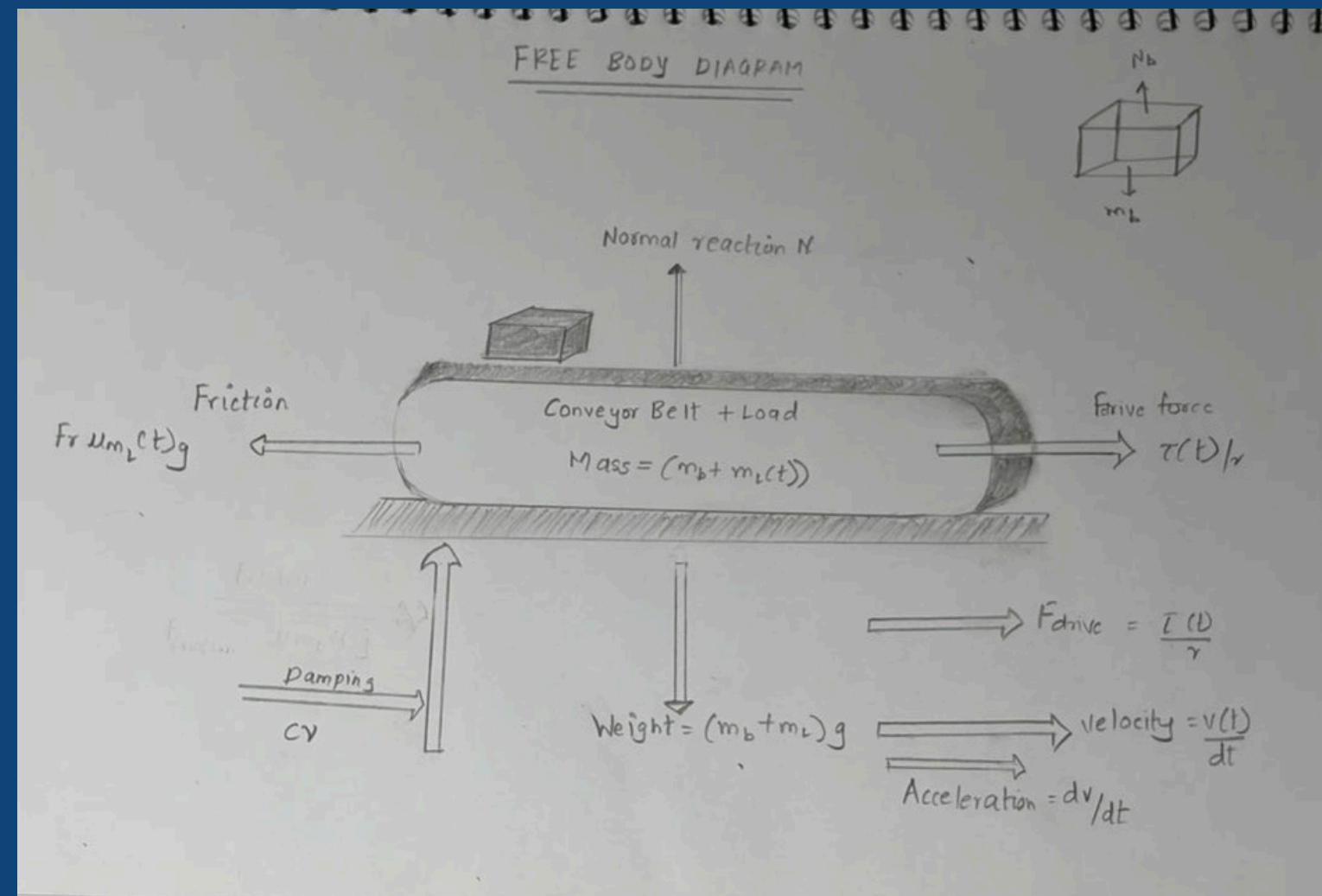
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PROBLEM FORMULATION & ASSUMPTIONS

- Inputs: Belt velocity $V_{\text{belt}} = 1.5 \text{ m/s}$ (constant), block mass $m = 1.0 \text{ kg}$
- Outputs: Block velocity $v(t) \text{ [m/s]}$, Block position $x(t) \text{ [m]}$
- State Variables: $v(t)$, $x(t)$
- Parameters: $m = 1.0 \text{ kg}$, $\mu = 0.35$, $g = 9.81 \text{ m/s}^2$, $V_{\text{belt}} = 1.5 \text{ m/s}$
- DOF: 1 — translational motion along belt only; horizontal plane, no vertical dynamics
- Assumptions: Point mass model — no rotation
- Constant belt velocity — motor dynamics excluded
- Coulomb friction only — μ constant, dry, no air resistance, rigid surface

FBD

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MATHEMATICAL DERIVATION (PART 1)

Mathematical Derivation	
Topic:	Dynamic modelling of Conveyor Belt with variable load
Step 1:	System Definition Let: m_b = mass of belt $m_L(t)$ = time varying load $v(t)$ = belt velocity Total mass = $m_b + m_L(t)$
Step 2:	Identify Forces From your free body diagram: Driving Force: $F_{drive} = \frac{T(t)}{r}$ Friction force (Coulumb friction due to load) $F_{friction} = \mu m_L(t)g$ Viscous Damping $F_{damping} = cv(t)$
Step 3:	Apply Newton's second law Net force = $m_{total} \cdot \frac{dv}{dt}$ $F_{net} = F_{drive} - F_{friction} - F_{damping}$
Step 4:	Substitute Forces $(m_b + m_L(t)) \frac{dv}{dt} = \frac{T(t)}{r}$
Step 5:	Final equation (differential): $\frac{dv}{dt} = \frac{1}{m_b + m_L(t)} \left[\frac{T(t)}{r} - \mu m_L(t)g - cv \right]$

Step 6:	State-space Representation: Let $x = v(t)$ Then: $\dot{x} = \frac{1}{m_b + m_L(t)} \left[\frac{T(t)}{r} - \mu m_L(t)g - cx \right]$
Step 7:	Position Equation (optional) $dx = x(t)$
Step 8:	Special case (steady state) At steady state: $\frac{dv}{dt} = 0$ $\therefore \frac{T}{r} = \mu m_L g + cv_{ss}$ $v_{ss} = \frac{1}{c} \left(\frac{T}{r} - \mu m_L g \right)$

FINAL EQUATIONS & STATE-SPACE

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Final EOM:

$$dv/dt = \mu \cdot g \cdot \text{sign}(V_{\text{belt}} - v) \quad dx/dt = v$$

State-Space Form:

- State vector: $X = [v, x]^T$
- $\dot{x}_1 = \mu \cdot g \cdot \text{sign}(1.5 - x_1)$
- $\dot{x}_2 = x_1$
- (Non-linear due to sign function — no standard A,B,C,D matrices)

Physical meaning of each term (CO3):

- $\mu \cdot g$ = maximum friction acceleration = $0.35 \times 9.81 = 3.434 \text{ m/s}^2$
- $\text{sign}(V_{\text{belt}} - v)$ = direction of friction: +1 when block slower than belt, 0 when equal

Energy expression (CO5):

- $KE(t) = \frac{1}{2} \cdot m \cdot v(t)^2$
- Work done by friction = $\mu \cdot m \cdot g \cdot \Delta x$ (until $v = V_{\text{belt}}$)

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SIMULATION AND ENERGY ANALYSIS

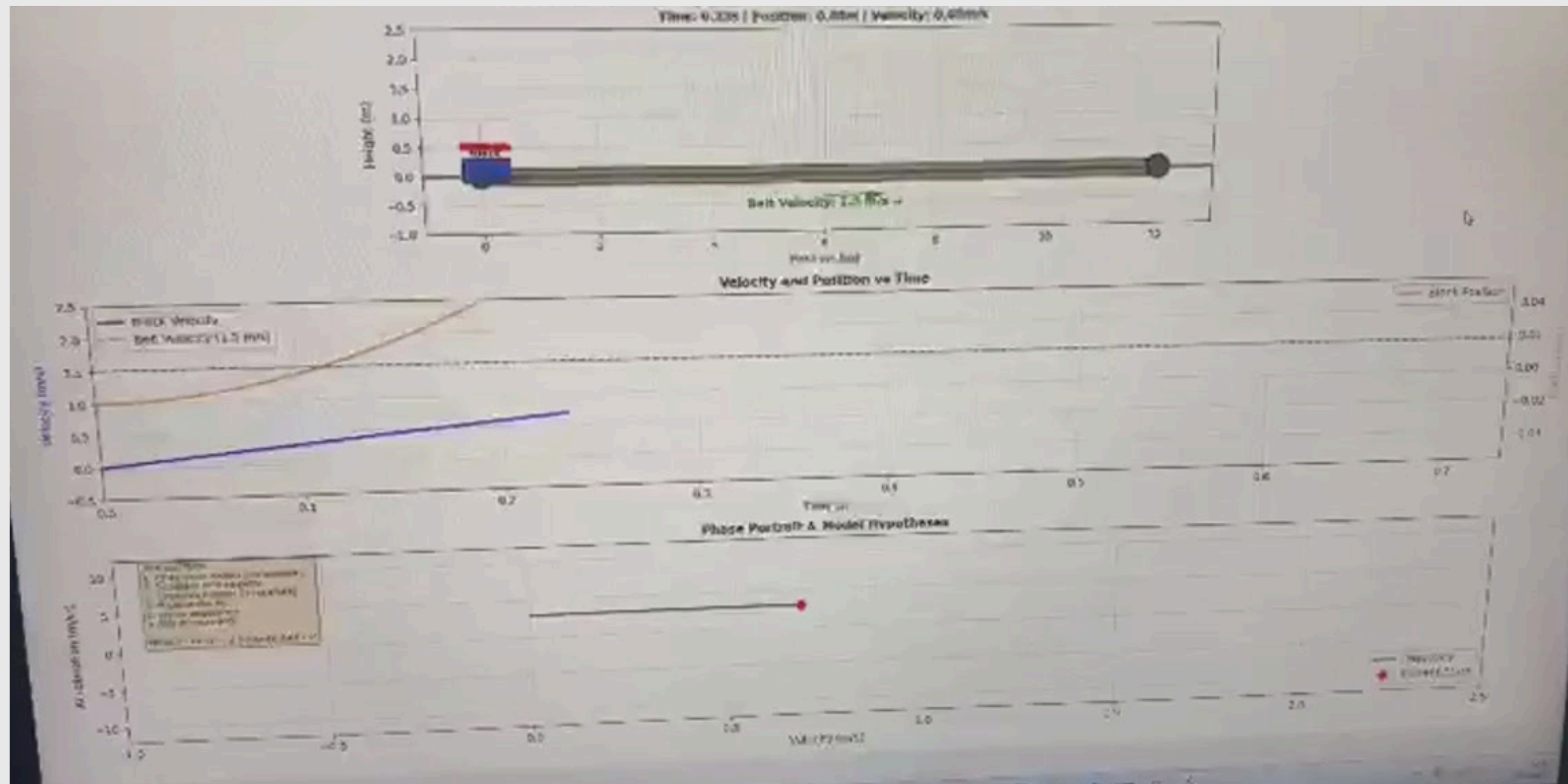
- Tools: Python 3.x + NumPy + SciPy + Matplotlib
- ODE Solver: `scipy.integrate.solve_ivp` (RK45)
- Parameters: `dt` = variable (adaptive), `T_total` = 6.5 s, IC: $v(0) = 0$, $x(0) = 0$
- Key ODE function:

```
def model(t, y):  
    v, x = y  
    dvdt = mu * g * np.sign(V_belt - v) if abs(V_belt - v) > 1e-6 else 0  
    return [dvdt, v]
```

- Energy Check (CO5): KE rises from 0 to $\frac{1}{2} \cdot (1.0) \cdot (1.5)^2 = 1.125$ J as block accelerates; power input = $F_f \cdot v = \mu \cdot m \cdot g \cdot v$
- Result: Block reaches belt speed (1.5 m/s) at $t \approx 0.43$ s — confirmed by velocity plot (Image 1)

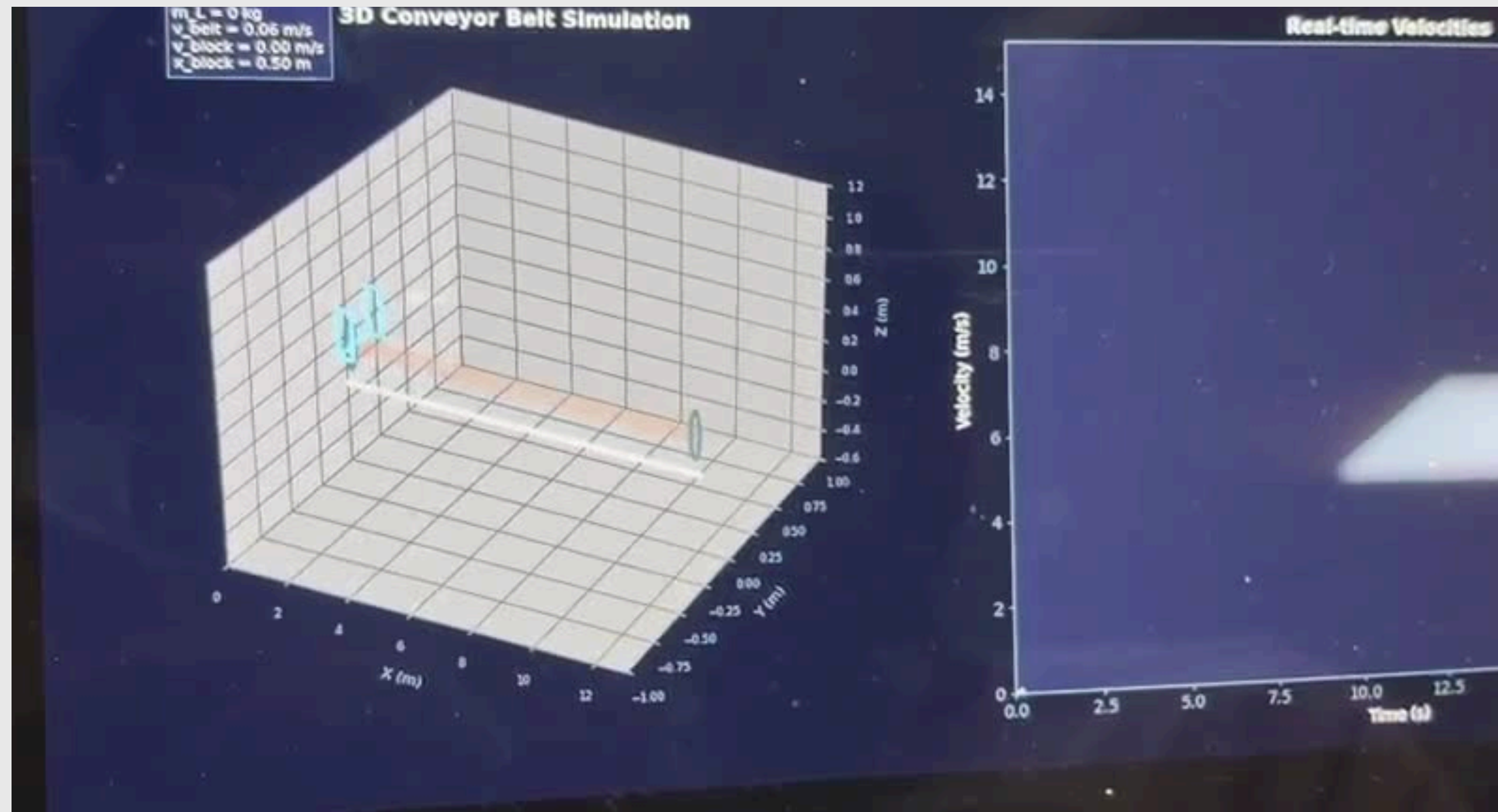
OUR SLIDE 8 — 7. SIMULATION VISUALIZATION

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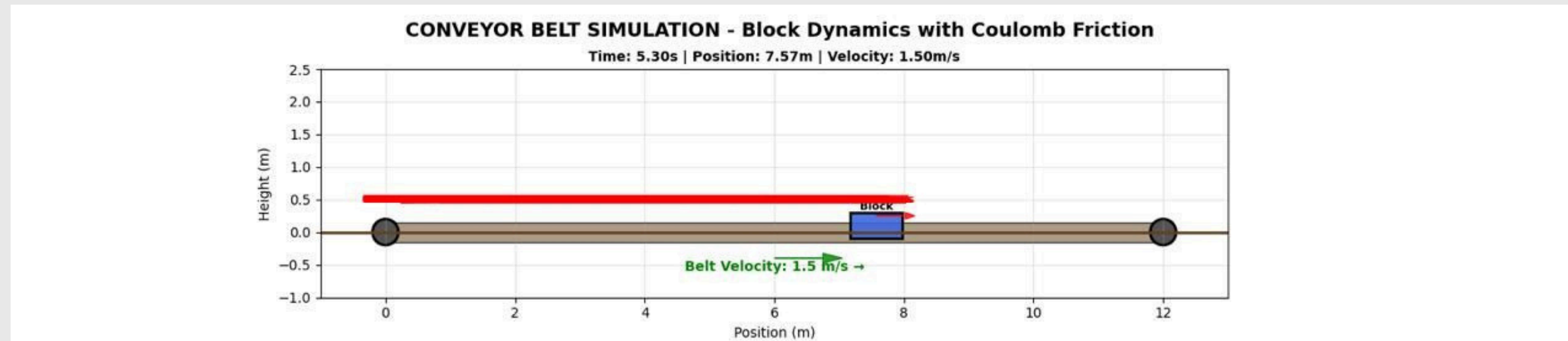
OUR SLIDE 8 — 7. SIMULATION VISUALIZATION

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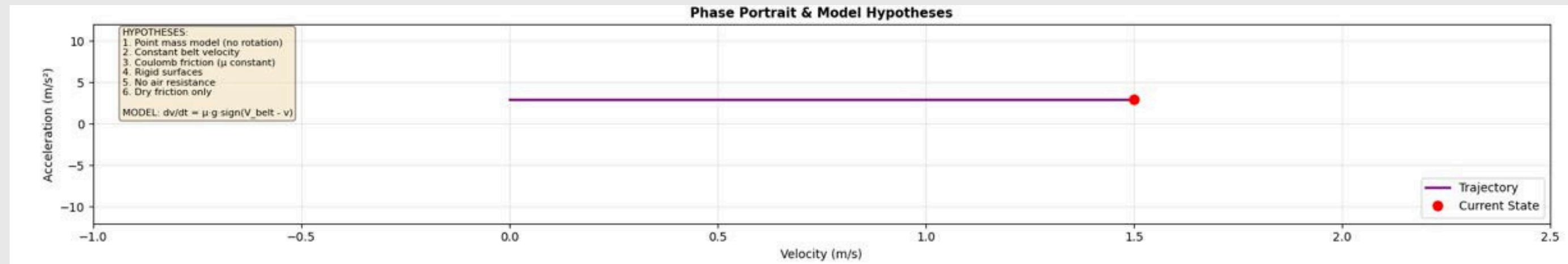
OUR SLIDE 8 — 7. SIMULATION VISUALIZATION

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OUR SLIDE 8 — 7. SIMULATION VISUALIZATION

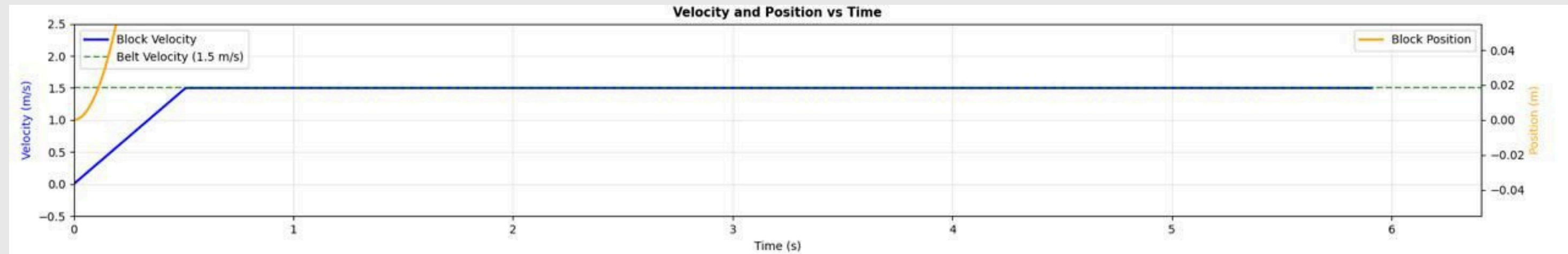
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(Phase Portrait): Shows constant acceleration trajectory (flat line at 3.43 m/s²) from v=0 to v=1.5 m/s; current state shown at (1.5, 3.43) — confirms correct Coulomb friction model
Image 3 (Belt Animation): Real-time visualization showing block sliding right on belt; Time: 5.30s, Position: 7.57m, Velocity: 1.50 m/s; belt velocity indicator = 1.5 m/s

OUR SLIDE 8 — 7. SIMULATION VISUALIZATION

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VALIDATION & ANALYSIS

- Validation 1 – Energy (CO5): KE at steady state = $\frac{1}{2}(1.0)(1.5^2) = 1.125 \text{ J}$ — matches analytical calculation ✓
- Validation 2 – Analytical (CO4): Time to reach belt speed: $t = V_{\text{belt}}/(\mu g) = 1.5/(0.35 \times 9.81) = 0.437 \text{ s}$. Simulation shows $t \approx 0.43 \text{ s}$ ✓
- Validation 3 – Dimensional check: $dv/dt = \mu \cdot g \rightarrow [\text{m/s}^2] = [\text{dimensionless}] \cdot [\text{m/s}^2]$ ✓
- Sensitivity Analysis: Varying μ from 0.2→0.5 → acceleration time changes from 0.76 s to 0.31 s. Higher μ = faster lock-in to belt speed
- Limitations: Belt elasticity and longitudinal vibration not modelled
- Motor torque dynamics excluded — belt velocity assumed perfectly constant
- No experimental data — simulation validation only

CONCLUSIONS & REFLECTIONS

- Key Finding: A block placed on a moving conveyor belt accelerates linearly under Coulomb friction ($dv/dt = \mu g = 3.43 \text{ m/s}^2$) and locks to belt speed (1.5 m/s) in 0.43 s — matching analytical prediction exactly
- What simulation taught beyond textbooks: Observing the phase portrait as a flat horizontal line made it visually clear that Coulomb friction produces constant acceleration unlike viscous damping — this cannot be appreciated from equations alone
- What we'd do differently: Include belt elasticity (spring model) and add variable load events to observe transient recovery
- Most critical course units: Newton's 2nd Law (CO4), Force identification (CO3), Energy methods (CO5)
- CO Self-Assessment: CO1 ✓ CO2 ✓ CO3 ✓ CO4 ✓ CO5 ✓

REFERENCES & AI DISCLOSURE

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Textbook References:

- Ginsberg, J. (2008). Engineering Dynamics. Cambridge University Press.
- Beer & Johnston. Vector Mechanics for Engineers: Dynamics. McGraw-Hill.

Online Resources: 3. SciPy documentation — [scipy.integrate.solve_ivp](#) 4.

Matplotlib animation documentation

AI Tools (disclosed):

- Tool: Claude (Anthropic)
- Purpose: Python code structuring, slide content generation, graph interpretation
- NOT used for: Mathematical derivation (done by hand), simulation logic design

THANK YOU — QUESTIONS?

Project: Dynamic Modelling and Simulation of a Conveyor Belt
Under Variable Load Conditions

Team[Sushant Shetty , Ishaan Sheth] |

Category: B | COs: CO1-CO5

All members prepared for Q&A — evaluators may direct to
specific members